Biological oxidation

The fermentation of the food consumed by an animal produces alcohol. To detoxify the alcohol, the liver produces an enzyme called alcohol dehydrogenase (ADH). Nicotinamide adenine dinucleotide (NAD) present in the animals act as an oxidising agent and ADH catalyses the oxidation of toxic alcohols into non-toxic aldehyde.

$$\underset{\text{ethanol}}{CH_3CH_2OH} + NAD^+ \xrightarrow{ADH} \underset{\text{ethanal}}{CH_3CHO} + NADH + H^+$$

Catalytic dehydrogenation

When the vapours of a primary or a secondary alcohol are passed over heated copper at 573K, dehydrogenation takes place to form aldehyde or ketone.

$$\underset{\text{ethanol}}{CH_3 - CH_2 - OH} \xrightarrow[573\ K]{Cu} \underset{\text{ethanal}}{CH_3 - CHO} + H_2$$

$$\underset{\text{Propan - 2- ol}}{CH_3-\underset{\displaystyle OH}{\underset{|}{CH}}-CH_3} \xrightarrow[573\ K]{Cu} \underset{\text{propanone}}{CH_3-\underset{\displaystyle O}{\underset{\|}{C}}-CH_3} + H_2$$

Tertiary alcohols undergo dehydration reaction to give alkenes.

$$\underset{\text{2 - methylpropan - 2- ol}}{CH_3-\underset{\displaystyle CH_3}{\underset{|}{\overset{\displaystyle CH_3}{\overset{|}{C}}}}-OH} \xrightarrow[573\ K]{Cu} \underset{\text{2 - methylprop - 1- ene}}{\begin{matrix} H_3C \\ \diagdown \\ \quad C = CH_2 \\ \diagup \\ H_3C \end{matrix}} + H_2O$$

Esterification

Alcohols react with carboxylic acids in the presence of an acid to give esters

Example

$$\underset{\text{ethanol}}{CH_3-CH_2-OH} + \underset{\text{ethanoicacid}}{HO-\overset{\displaystyle O}{\overset{\|}{C}}-CH_3} \xrightarrow{H^+} \underset{\text{ethylethanoate}}{CH_3-\overset{\displaystyle O}{\overset{\|}{C}}-O-CH_2-CH_3}$$

XII U11-Hydroxy compounds.indd 118 11-01-2021 15:44:49